

Confirmatory Factor Models

Identification

Agenda

- Revisiting identification in the latent variable context
- Scaling constraints
- Identification Rules
- Empirical tests of local identification

What is identification?

If an unknown parameter in θ can be written as a function of one or more elements of Σ , that parameter is identified. If all unknown parameters in θ are identified, then the model is identified.

This definition emphasizes how we can actually go about testing whether a model is identified.

Consider two potential parameter vectors: θ_1 and θ_2 . The implied covariance structures for these two parameter vectors are $\Sigma(\theta_1)$ and $\Sigma(\theta_2)$. The model is identified if when $\Sigma(\theta_1) = \Sigma(\theta_2)$ means that $\theta_1 = \theta_2$.

No two parameter vectors result in the same implied covariance unless they are equal.

What's in θ ?

Let's say we have N_Z observed variables and N_L latent factors

In a confirmatory factor model there are three parameter matrices of interest:

$\Lambda_{N_Z \times N_L}$ this will have $N_Z \times N_L$ unique values (with no restrictions)

$\Sigma_{\epsilon_L \epsilon_L}_{N_L \times N_L}$ this will have $\frac{1}{2}(N_L)(N_L + 1)$ unique values (with no restrictions)

$\Sigma_{\epsilon_Z \epsilon_Z}_{N_Z \times N_Z}$ this will have $\frac{1}{2}(N_Z)(N_Z + 1)$ unique values (with no restrictions)

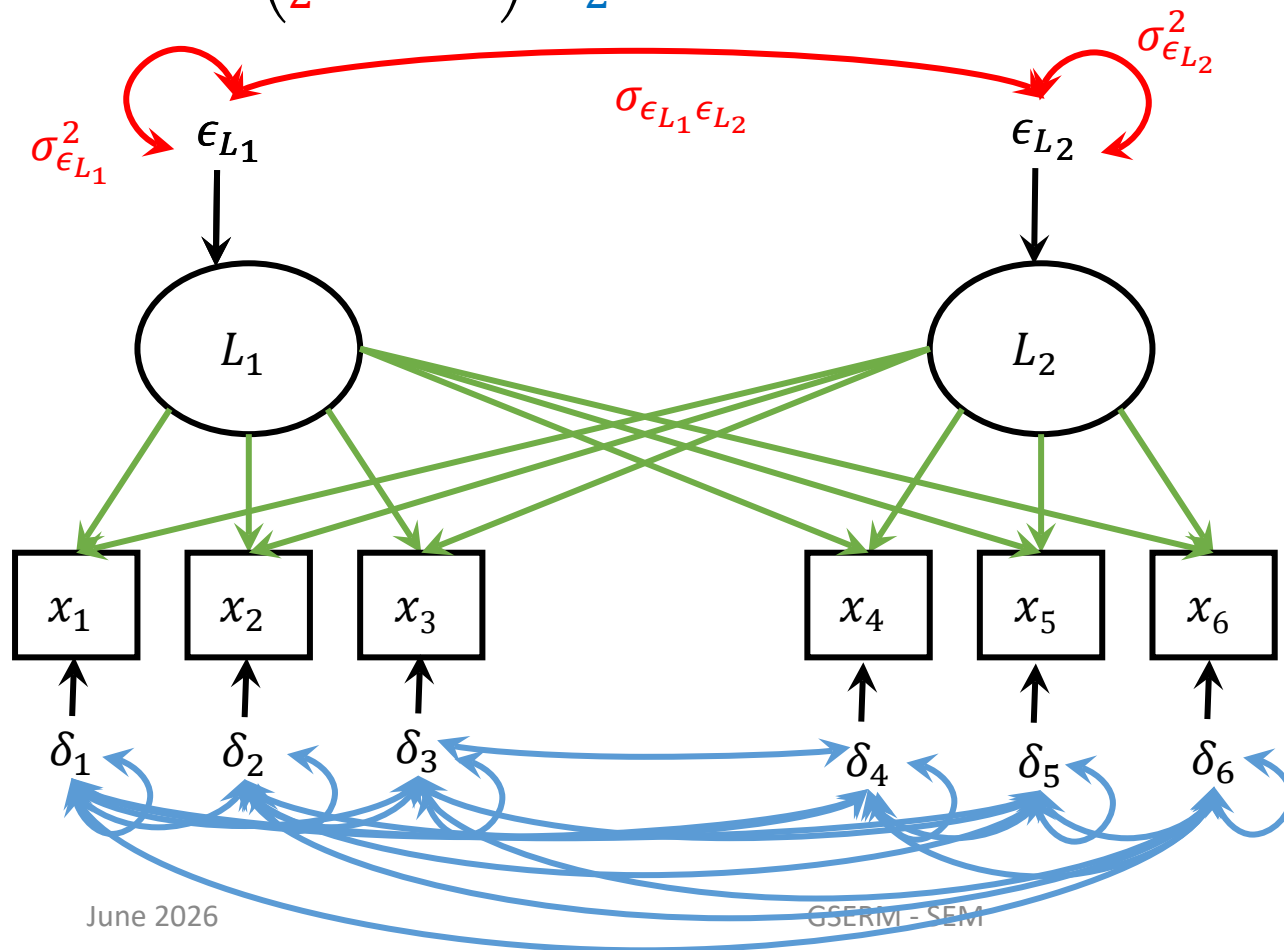
In total a freely estimated model would have:

$$N_\theta = N_Z \times N_L + \left(\frac{1}{2} N_L (N_L + 1) \right) + \frac{1}{2} (N_Z (N_Z + 1))$$

A model under No constraints

To try to estimate this information we have $\Sigma_{N_Z \times N_Z}$ which has $\frac{1}{2}(N_Z)(N_Z + 1)$ unique values.

$$6 \times 2 + \left(\frac{1}{2} 2(2 + 1) \right) + \frac{1}{2} (6(6 + 1)) = 36$$



$$\frac{1}{2} (6)(6 + 1) = 21$$

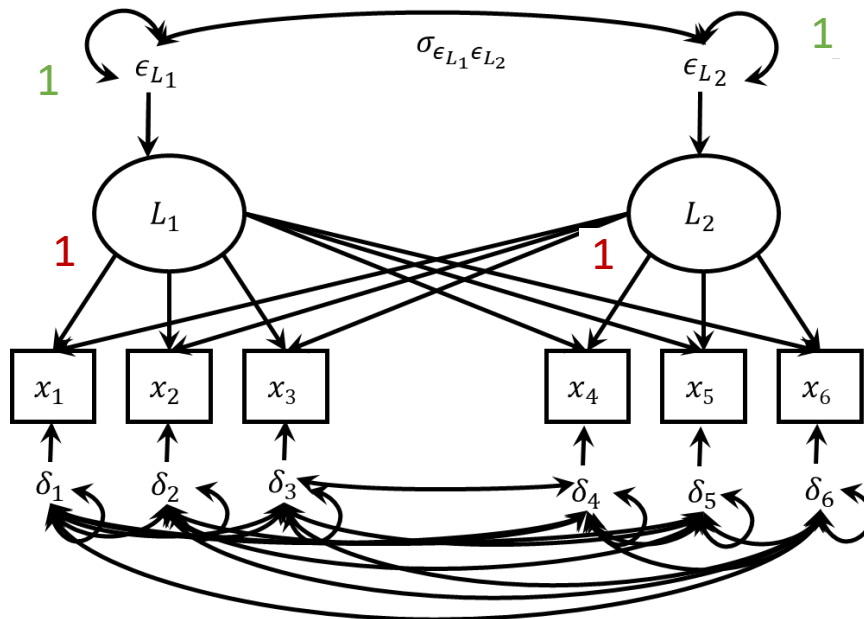
By the N_θ -rule this model is not identified

Scaling Constraints

Latent variables do not often have a natural scale, so there are two sets of constraints commonly used:

1. Constrain the first loading of the first item on each latent factor to 1. Scale of latent variable is equal to scale of first item
2. Constraint the variance of each latent factor to 1. “Standardized” latent variable.

These constraints are used to set the scale of the latent variables, but also get us 1 step toward identification.



Either choice will reduce the number of estimated parameters by N_L

Common Constraints on $\Sigma_{\epsilon_Z \epsilon_Z}$

$\Sigma_{\epsilon_Z \epsilon_Z}$ is the variance covariance matrix of the deviations for the observed variables.

Common factor models typically propose that the factor is the **only reason for shared variance** among the observed variables.

This means that for many cases, there will be few or no covariances between deviances.

It is common to assume that $\Sigma_{\epsilon_Z \epsilon_Z}$ is **diagonal**.

$$\Sigma_{\epsilon_Z \epsilon_Z} = \begin{bmatrix} \sigma_{\epsilon_{Z_1}}^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & \sigma_{\epsilon_{Z_2}}^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sigma_{\epsilon_{Z_3}}^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_{\epsilon_{Z_4}}^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma_{\epsilon_{Z_5}}^2 & 0 \\ 0 & 0 & 0 & 0 & 0 & \sigma_{\epsilon_6}^2 \end{bmatrix}$$

This was described as part of the “typical” CFA last time

This constraint reduces the estimated parameters by $\frac{1}{2}N_Z(N_Z - 1)$

Common Constraints on Λ

A common constraint for the Λ matrix is that only one element of each row is freely estimated.

This is equivalent to having zero cross-loadings (i.e., each item loads on one and only one latent factor)

but with scaling constraints
it would be...

$$\Lambda = \begin{bmatrix} 1 & 0 \\ \lambda_{21} & 0 \\ \lambda_{31} & 0 \\ 0 & 1 \\ 0 & \lambda_{52} \\ 0 & \lambda_{62} \end{bmatrix}$$

This was described as part of the “typical” CFA last time

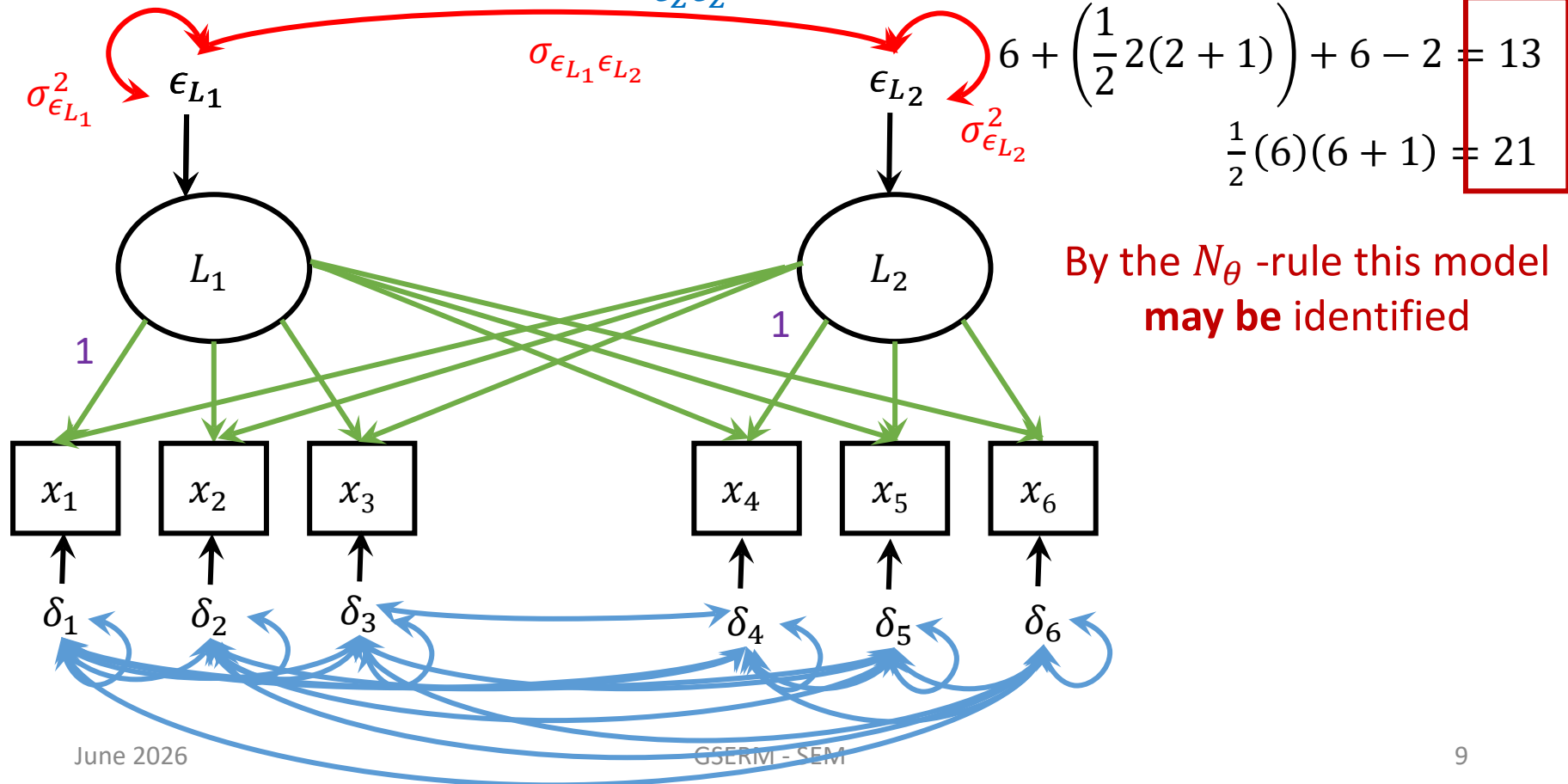
This constraint reduces the estimated parameters by $N_Z(N_L - 1)$

What information do we have?

Under no constraints

$$N_{\theta} = N_Z + \left(\frac{1}{2} N_L (N_L + 1) \right) + N_Z - N_L = 2N_Z + \frac{1}{2} N_L (N_L - 1)$$

But with scaling constraints, assuming $\Sigma_{\epsilon_Z \epsilon_Z}$ is diagonal, assuming unidimensionality in Δ



Identification Rules

N_θ -rule

The N_θ -rule applies to CFA models as well as observed variable models.

The N_θ -Rule is *necessary but not sufficient* for identification

Remember N_θ is the length of θ , a list of all non-repeating and freely estimated parameters.

If $N_\theta \leq \frac{1}{2}(N_Z)(N_Z + 1) \dots$

The model **may** be identified

If $N_\theta > \frac{1}{2}(N_Z)(N_Z + 1) \dots$

The model **is not** identified.

Three-Indicator Rule

This rule is a **sufficient but not necessary** condition for identification

It applies to...

1 factor models only

Conditions:

- At least 3 indicators with non-zero loadings
- Σ_{ϵ_Z} is diagonal

Multifactor version:

- Three or more indicators per latent variables
- Each row of Λ has one and only one non-zero element
- Σ_{ϵ_Z} is diagonal

Two-Indicator Rule

This rule is a **sufficient but not necessary** condition for identification

It applies to...

models with more than 2 factors

Conditions:

- Latent Variables must be scaled (i.e., loading set to 1, or fixed variance)
- At least 2 indicators per latent factor with non-zero loadings
- Σ_{ϵ_Z} is diagonal
- No zero elements in Σ_{ϵ_L}

A more general approach for Σ_{ϵ_L}

- Each row of Λ has one and only one non-zero element
- Each row of Σ_{ϵ_L} has at least 1 nonzero off-diagonal element
- Σ_{ϵ_Z} is diagonal

Empirical Tests of Identification

Limitations of Rules

- The identification rules do not provide a complete method for determining identification for all CFA models
 - Correlated residuals are typically not allowed
 - Cross-loadings are often not allowed
- In these cases researchers may turn to empirical tests of identification

Global and Local Identification

- **Global Identification:** This is the definition of identification we've been working with.

Consider two potential parameter vectors: θ_1 and θ_2 . The implied covariance structures for these two parameter vectors are $\Sigma(\theta_1)$ and $\Sigma(\theta_2)$. The model is identified if when $\Sigma(\theta_1) = \Sigma(\theta_2)$ means that $\theta_1 = \theta_2$.

Applies to all values of θ

- **Local Identification:** This is a new concept

A parameter vector θ is locally identified at the point θ_1 , if in the neighborhood of θ_1 there is no vector θ_2 for which

$\Sigma(\theta_1) = \Sigma(\theta_2)$ unless $\theta_1 = \theta_2$

Applies to a specific point along θ

Global identification implies local identification

Local identification is necessary but not sufficient for global identification

Wald's Rank Rule

$\Sigma(\theta)$ is the model implied variance covariance matrix

new notation! $\sigma(\theta)$ is a vector of all the unique elements from $\Sigma(\theta)$

We can take the partial derivative of each element of $\sigma(\theta)$ by each element of θ

$$\frac{\partial \sigma(\theta)}{\partial \theta}$$

This results in a matrix

If this matrix has a rank of t when evaluated at θ_1 the model is locally identified at θ_1

Keesling's Test of Local Identification

Information matrix:

$$-E \left(\frac{d^2 F_{ML}}{d\boldsymbol{\theta} d\boldsymbol{\theta}'} \right)$$

If this matrix is invertible then $\boldsymbol{\theta}$ is locally identified at $\boldsymbol{\theta}_1$

Although the calculations are not specific to $\boldsymbol{\theta}_1$ this is **not a test of global identification**

The inverse of the information matrix is calculated by most software (lavaan included) to calculate the standard errors.

If you get solutions with no standard errors this can suggest your model is not identified.

A note of warning:

1. Computers are imperfect at taking inverses
2. We are checking identification at $\hat{\boldsymbol{\theta}}$ (our estimated parameters) rather than $\boldsymbol{\theta}$ (the true parameters)